

Stock Arena

A daily stock-picking contest.

Real market prices, fake money, and a contest that starts and finishes inside a single trading day. This draft replaces the weekly league of Draft v1 — because the thing most likely to kill this app was never the market data, the 3D chests, or the choice of builder platform. It was a *five-day feedback loop on an asset class that moves one percent a day.*

REVISION

DRAFT V2

SUPERSEDES

DRAFT V1

SCOPE

V1 RELEASE

STATUS

FOR DECISION

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What changed from Draft v1

Four decisions separate this draft from the last one. Everything else in the document follows from them.

CHANGE 01 Daily contests, not weekly leagues A week is too slow a loop for a mobile app — by Tuesday most players know they cannot win and stop opening it. A daily contest is also simpler to build: no mid-week joins, no multi-day state, no partial settlement. And you learn whether it is fun in a day instead of five.	CHANGE 02 Engagement mechanics are the product Draft v1 treated drafts, duels, feeds, and progression as polish. They are the reason anyone opens the app twice. Shipping the trading engine without them tests nothing worth knowing, so they are all in v1.
CHANGE 03 A real reward system "Coins and chests" was a placeholder. It is replaced by a Brawl Stars-style trophy ladder with a Trophy Road, per-ticker ranks, and two currencies — designed so that losing costs you something, which is what makes winning mean anything.	CHANGE 04 Tool comparison removed It was 70% of Draft v1 and covered the most reversible decision in the project. It is replaced by a single architectural requirement — you need a scheduled job and a real database — which is the actual filter to run platforms through.

The three problems Draft v1 did not see

PROBLEM	WHY IT MATTERED	RESOLVED IN
Delayed prices with instant fills	Filling orders at a 15-minute-old price is a free-money exploit — anyone with a live quote elsewhere buys into a move that already happened. Every player who noticed would win; every player who did not would quit.	\$6 — forward pricing
Data licensing	Vendor choice was framed as a rate-limit question. The real question is whether you may redistribute the data to users at all. Free tiers are typically personal, non-commercial use.	\$9 — hard constraints
Chests are a loot box	Randomized rewards bought with an earned currency require published odds under App Store rules and are regulated in several jurisdictions. That is a review-blocking workstream, not a polish item.	\$5 — deterministic reveals

Stocks are slow. A big day is two percent. You cannot fix that, and every attempt to make the price itself feel exciting will fail against a user who has played a mobile game before.

So do not compete on price movement. Compete on **rank**. Rank is zero-sum, discrete, and churns constantly — in a room of thirty players, a 0.3% move reshuffles five positions. That is a real event, it happens all day, and it costs nothing extra to produce.

Never show "+0.41%". Show "↑3 spots — two behind Marcus."

Same underlying data, entirely different emotional product. Every design decision in this document follows from this one: shorten the contest, curate for volatility, score in points, and generate events from what players do to each other rather than from what the market does.

What this buys you

SLOW-DATA PROBLEM	THE RANK-FIRST ANSWER
Nothing happens for hours	Rank churns continuously even on a flat tape. The room feed always has something to report.
Small numbers feel small	Score in points — 1% equals 100 points. "You gained 340 points" reads as an achievement; "+0.34%" reads as a rounding error.
Market is closed 17 hours a day	Closure becomes ritual. The pool drops at 8 PM, the draft runs overnight, and the overnight gap is the biggest single move most stocks make.
Losing players disengage	Trophies, ticker ranks, and Trophy Road progress accrue on a different axis from any single day's result.

Core loop

The whole product resolves inside a single trading day. A player who joins at 9 AM has a result by 4:05 PM, and a reason to come back tomorrow.

8:00 PM PREV. Pool drops Tomorrow's 20 curated tickers are published. Push notification.	8 PM – 9:30 AM Draft Snake-draft your book against the room. Exclusive ownership.	9:30 AM Bell Positions lock. Live feed opens. Rank starts moving.	ALL SESSION Five trades A hard budget of five. Every one is a real decision.	4:00 PM Buzzer Settlement at close. Final rank, trophies, graded recap.
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Why the day is shaped this way

The evening pool drop and overnight draft do most of the work. They move the moment of highest tension — deciding what to own — *outside* market hours, where you control the clock entirely and are not dependent on anything moving. By the time the bell rings, every player has already made their most consequential decision and has something at stake.

The five-trade budget then keeps players engaged during the session without letting them churn. Scarcity is what turns a trade from a tap into a decision worth agonizing over — and it incidentally caps your write load and API pressure.

What ships in v1

Everything here is in scope for the first release.

FORMAT The daily contest One contest per trading day, opening at 8 PM the night before and settling at the close.	UNIVERSE Curated 20-ticker pool Hand-picked high-beta and news-driven names, rotated daily. Two problems solved at once: real movement, and no paralysis in front of a 5,000-symbol search box. The rotating pool also gives the game a meta that changes.
ENTRY Snake draft, exclusive ownership Draft your book against the room. If you take NVDA, nobody else in your room can. All the tension lands before the market even opens, on a schedule you control.	ECONOMY Five trades a day A hard action budget, no roll-over. Creates decisions, kills spam-trading, and caps API load.
SCORING Points, never percentages 1% equals 100 points. Same number, better game.	BALANCE 20% position cap Without a cap, the optimal play over one session is dumping everything into the single most volatile name — pure coin flip, zero skill, boring by week two. Caps force a real portfolio.
STRUCTURE Rooms of 30 Small rooms mean everyone is near the top of <i>something</i>. A global leaderboard with nine players on it is worse than no leaderboard at all.	STRUCTURE 1v1 duels Challenge a friend, 24 hours, higher score wins. Emotionally far stronger than "17th of 30" — and it needs two players instead of thirty, which solves the cold-start problem outright on launch day.
PROGRESSION Trophies & Trophy Road Placement moves your trophy count up or down; milestones on a single linear road unlock features and cosmetics. Full spec in §5.	IDENTITY Track Record A permanent career equity curve. Every contest — win or lose — adds a point to one lifetime chart that never resets. The reward for playing a trading game should be a trading record.

SIGNAL Ownership % "Only 6% of your room holds this." Turns a flat tape into a bet against specific people, and gives contrarian players an identity to play toward.	RISK Call Your Shot Before the bell, spend Gold to publicly declare one pick to your room. If it finishes as the day's top performer, big multiplier and a permanent mark on your record. If not, everyone watched you miss.
EVENTS Live room feed Sports-broadcast framing for slow data: 10:42 – Sarah dumped SOFI, up 4 spots. Your feed does not have to be as slow as the market.	EVENTS Notifications with stakes "You passed 4 players." "One spot from the payout." "Earnings in 20 minutes." The highest-leverage retention feature in the build, and it was missing from Draft v1 entirely.
PAYOFF Graded recap A 15-second animated replay of your book against the room, plus a grade: <i>"B+. You sold NVDA 40 minutes early — cost you 3 spots."</i> Shareable, and the honest version of what the buddy was always meant to deliver.	PAYOFF 3D reward reveal The spinning-open chest animation from Draft v1, attached to Trophy Road milestones instead of randomized boxes. Same moment, none of the compliance cost.

Modelled directly on Brawl Stars, because the structure fits this game better than it has any right to. In Brawl Stars every character holds its own trophies and your total is the sum of them. Here, *every ticker is a brawler*.

Finish a session green holding NVDA and NVDA gains trophies. Your total is every ticker you have ever played, added together. That single decision turns the rotating daily pool into "which of my brawlers are playable today," and gives you a progression system with no authored content in it at all — the market supplies the roster.

Trophies go down as well as up.

This is what a softer, everyone-gets-something economy gets wrong. If finishing 28th costs nothing, finishing 1st means nothing. Losses have to be visible on the ladder for wins to carry weight — and the drop stays small enough that a bad day is a setback, never a wipeout.

Trophy deltas

FINISH, ROOM OF 30	TROPHIES	EFFECT
1st	+12	A real climb — several days of drift undone in one session.
2nd – 3rd	+9	The podium is worth chasing right up to the buzzer.
4th – 6th	+7	
7th – 10th	+5	
11th – 15th	+2	Top half always pays something.
16th – 20th	0	A neutral band, so an average day is not a punishment.
21st – 25th	-3	
26th – 30th	-5	Roughly half a win. Enough to sting, not enough to quit over.

Apply the Brawl Stars soft cap on top: past roughly 1,000 trophies, gains shrink and losses grow with each bracket. The climb has to get harder or the ladder is just a

clock, and high-trophy rooms stop meaning anything.

Trophy Road

One linear track, visible from day one, so a new player can always see the next thing they are playing toward. Note how much of it is **features** rather than decoration — the game keeps opening up for months.

TROPHIES	LEAGUE	UNLOCKS
25	—	Shop opens · name colors
50	Bronze	1v1 duels
150	—	Signature ticker badge goes live
250	Silver	Private rooms — invite friends
600	Gold	Bracket tournaments
800	—	Trophy case — three display slots
1,000	Diamond	Draft entrance animations
1,500	—	Custom pool contests — pick your own 20
2,000	Mythic	Second Call Your Shot slot
3,000	—	Earnings-night contests
5,000	Legendary	Animated card frame
10,000	Masters	Global leaderboard placement

Ticker ranks

Each ticker ranks up on its own trophies, exactly like a brawler. Rank 5, 10, and 20 award that ticker's bronze, silver, and gold badge — and a gold-ranked ticker can be worn as your **signature** next to your name in every room. Marcus · NVDA

The effect is that players build a *stable*, and an identity from it. Someone becomes a semis player and starts caring which stocks are in tomorrow's pool for reasons that have nothing to do with today's contest.

Two currencies, and no third

COMMON Gold Earned from Trophy Road milestones and daily quests. Buys shop cosmetics and Call Your Shot declarations. Flows steadily, so the shop is where a losing week still produces a win.	RARE Star Points Earned only at the monthly season reset, scaled by how far above the reset threshold you climbed. Buys prestige items nothing else can — the reason a Legendary player keeps pushing past a rank they have already secured.
NOT IN V1 No purchasable currency Nothing is bought with money and nothing converts to money in either direction. Keeping this true is what keeps the app a game rather than a contest with prizes of value. See §9.	RULE Never gate information or edge No unlocking better stats, faster prices, extra trades, or a larger position cap. The moment progression makes a veteran better at the game rather than more visible in it, you have built pay-to-win and new players quit.

The shop

One hard rule: **only sell things other players in your room can see**. A skin nobody sees has no value; a badge thirty people see every morning does. Name colors, leaderboard badges, card frames, draft entrance animations, Track Record chart skins, and earned *titles* — a string next to your name like *Diamond Hands* or *The Fade*, which cost nothing to author and carry enormous social weight.

Stock rotates daily, three items at a time. Same appointment mechanic as a chest, same 3D reveal, none of the randomness.

Season reset

Monthly. Everyone above the reset threshold is pushed back toward it and converts the difference into Star Points. Peak league is kept permanently on your card, so the climb is never erased — only the ladder is. This stops the top of the game calcifying and gives every month a fresh start for players who fell off.

Simulation spec

Draft v1 left all of this unspecified. Several of these are not preferences — they are the difference between a game and an exploit.

RULE	V1 DECISION	WHY
Fill price	Forward pricing. An order placed at time T fills at the <i>next</i> price the system publishes after T.	Filling at the last published price is a free-money exploit: anyone with a live quote elsewhere buys into a move that already happened. Forward pricing closes it completely and costs nothing.
Price source	Delayed data, cached server-side, refreshed on a fixed interval.	With forward pricing, delay stops mattering for fairness — and delayed data is dramatically simpler to license than real-time.
Position cap	20% of book per holding. Minimum four positions.	Kills the max-variance degenerate strategy.
Trade budget	Five per session. Unused trades do not roll over.	Scarcity creates decisions; also caps write load and API pressure.
Shorting	Not in v1. Long-only.	Doubles the edge cases — borrow, unlimited loss, forced close — for no proven engagement gain.
Fractional shares	Yes. Allocate by percentage of book, not share count.	Removes an entire class of rounding bugs and makes a \$500 stock as playable as a \$5 one.
Halts & missing quotes	Symbol freezes at last good price; trades on it are rejected with a visible reason.	Silent failures here read as cheating to the player who lost the spot.
Settlement	Official close, 4:00 PM ET. Half-days honored via a market calendar. Job is idempotent and keyed by contest ID.	A settlement cron that fires twice and double-pays is the single most damaging bug in the system.
Ties	Broken by fewest trades used, then by earliest draft entry.	Rewards conviction, and is deterministic.
Late joins	Locked out once the bell rings. Next contest opens at 8 PM.	Cleaner than scaling a partial entry, and it makes the evening pool drop matter.

One rule about market data, and it determines almost everything else.

Clients never touch the data vendor.

A server-side job polls the union of symbols in today's pool — twenty of them, not five thousand — writes prices to your own database on a fixed interval, and every client in every room reads that cache.

This is what makes a 60-call-per-minute free tier survivable. Draft v1 called that limit "generous," but 150 distinct held symbols would have consumed the entire quota on a single refresh, with no headroom, forever. The curated pool is therefore an **architectural** decision as much as a design one.

It also means the published price ticks on *your* schedule, which is what makes forward pricing implementable at all.

What this implies for tooling

You need a **scheduled background job** and a **real database**. Filter every platform in Draft v1's comparison table through that requirement before comparing anything else — most free builder tiers do not offer cron, which disqualifies them regardless of how many credits they hand out.

That comparison table is otherwise omitted from this draft on purpose. It covered the most reversible decision in the project at the greatest length, its numbers go stale within a month or two, and the real cost of running this app is hosting and market data — neither of which appeared in it.

Deliberately not in v1

Nothing below is abandoned except where marked. Each is either blocked on evidence the game is fun, or carries a compliance cost that should not be paid before that evidence exists.

Randomized chests CUT PERMANENTLY — REPLACED	Randomized rewards bought with an earned currency are a loot box: Apple requires published odds and several jurisdictions regulate them. Deterministic Trophy Road reveals deliver the same moment — including the 3D animation, which <i>is</i> in v1 — with none of the compliance cost. There is no version of this worth reintroducing.
The buddy V2 — STRONGEST CANDIDATE	Genuinely good retention design; a character that reacts when you are losing is a reason to open the app on a bad day. But the graded recap delivers the same insight payload in v1 for a fraction of the cost, and validates whether anyone reads it before you commission a rigged 3D character.
1K/ 10K/ 100K tiers RECONSIDER, DO NOT REBUILD	With percentage-based scoring the three tiers are mechanically identical — one league with three skins, splitting a small player pool three ways and making every leaderboard look empty. Cut until there is a real mechanical difference between them.
Weekly leagues V1.1 — AS A SEASON LAYER	Comes back as the season that daily contests roll up into, not as the primary format.
Brackets & earnings nights V1.1 — ON THE TROPHY ROAD	Both already sit on the Trophy Road as unlocks. Earnings nights and macro days — CPI, Fed, jobs — are free volatility on a published calendar. Build the engine so that a "contest" is simply a row with a start, an end, and a pool, and these cost almost nothing to add.
Crypto contests V1.1 — WITH LEGAL REVIEW	Trades 24/7 and moves far more than equities, which fills dead weekends and the overnight gap. Opens its own regulatory questions and deserves its own look before it ships.
Clans V1.1	The strongest retention mechanic in mobile gaming — teammates notice when you do not show up. But it needs a population to be anything other than empty.
Ticker card collection V2	A card per stock finished green, organized into sector sets. Overlaps heavily with ticker ranks; hold until you know whether players want a second collection layer.

Hard constraints

Non-negotiable. Most of these are legal, and all of them are cheaper to honor than to retrofit.

Trophies, Gold, and Star Points are never cashable, tradable, or purchasable

Write this into the plan as a permanent constraint, not a v1 simplification. A ranked contest with prizes of real value pulls in sweepstakes and skill-contest law across US states. Non-cashable, cosmetic-only rewards are the thing keeping this a game.

Confirm the data license covers redistribution

Free market-data tiers are typically personal, non-commercial use. Showing quotes to your users is redistribution. Real-time exchange data to end users generally requires agreements with the exchanges themselves; delayed data is far less encumbered. Verify before writing the integration, not after.

The unofficial Yahoo endpoint is not a foundation

Against Yahoo's terms, and you will design your data layer around its response shape and then have to rebuild it. Acceptable as a one-off cross-check on your own machine; not in the app.

No loot boxes, in any version

Deterministic reveals cost nothing extra and remove an entire compliance surface — published odds, jurisdictional bans, and App Store review risk. Once the animation is decoupled from the randomness there is no upside left to recover.

Nothing in the app is investment advice

The graded recap describes what happened in a simulation. It never suggests what to buy, and it never refers to positions outside the game.

Ordered so the riskiest assumption — *is a one-day contest fun?* — is testable as early as possible, and so nothing downstream is built on an unproven core.

01 FOUNDATION	Accounts. Database schema for contests, rooms, entries, positions, and price ticks. The scheduled price-cache job running against the curated pool, with the market calendar in place from day one. SHIPS: NOTHING VISIBLE. EVERYTHING DEPENDS ON IT.
02 ENGINE	Contest lifecycle: open, draft, lock, live, settle. Forward-priced fills, position caps, the trade budget, points scoring, and an idempotent settlement job. Test settlement against a replayed historical day before a single screen is designed. SHIPS: A CONTEST THAT RUNS CORRECTLY WITH NO UI.
03 PLAYABLE	Draft screen, portfolio view, trade flow, live room leaderboard. Rank-first presentation throughout — spots gained and lost, never raw percentages. SHIPS: THE FIRST INTERNAL PLAYTEST. STOP HERE AND DECIDE HONESTLY WHETHER IT IS FUN.
04 STAKES	Trophies and the Trophy Road, ticker ranks, 1v1 duels, the live room feed, ownership percentages, Call Your Shot, and push notifications. This is the phase that converts a working simulation into something anyone opens twice. SHIPS: THE FIRST BUILD WORTH PUTTING IN FRONT OF STRANGERS.
05 PAYOFF	Track Record, the animated recap and graded post-mortem, the shop and 3D reward reveal, sharing. Basic anti-cheat — multi-account detection, since duplicate accounts directly attack the trophy economy the moment there is anything to farm. SHIPS: V1.

11 Open decisions

What this plan deliberately does not settle, roughly in the order they need answering.

- Q1 Mobile-native or mobile web?** Draft v1 said "mobile app" but every tool it evaluated — Bolt, Lovable, Replit, v0, React Three Fiber — builds for the web. This determines whether App Store rules apply to you at all, and it needs answering before phase 3.
- Q2 Who curates the daily pool, and how?** Hand-picked is right for launch and does not scale. Decide what the rule becomes — top movers, highest implied volatility, the earnings calendar — before it turns into a daily chore.
- Q3 Draft or free-pick as the default entry?** Drafting is the stronger mechanic but demands a populated room. A free-pick fallback may be needed while numbers are thin.
- Q4 Where is the season reset threshold?** Too low and everyone gets pushed down every month, which feels punishing; too high and Star Points never flow. Start near 1,000 — where the soft cap begins — and tune once you can see the real distribution.
- Q5 How many trophies does a Trophy Road milestone award in Gold?** The whole shop economy hangs off this number and it cannot be set sensibly until phase 4 is playable.
- Q6 Monetization, if any.** Still unanswered, still the decision that determines how much compliance work exists. Cosmetics only is the low-risk path and preserves every constraint in §9.